

SECTION B

6 (short) Using Laplace transforms, solve the differential equation

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + x = 1$$

with initial conditions $x = dx/dt = 0$ at $t = 0$. [10]

7 (long)

(a) The response $y(t)$ of a linear system to a general input $f(t)$ is given by the convolution integral

$$y(t) = \int_0^t f(\tau) g(t - \tau) d\tau$$

where $g(t)$ is the impulse response. Show that this integral and

$$y(t) = \int_0^t g(\tau) f(t - \tau) d\tau$$

are equivalent. [3]

(b) A linear system with input $f(t)$ is governed by the equation

$$\alpha \frac{dy}{dt} + y = f$$

where α is a constant.

(i) Without using Laplace transforms, find the step response of the system, assuming that $y(t) = 0$ for $t < 0$. [8]

(ii) Find the impulse response $g(t)$ of the system. [4]

(iii) Find the response to an input

$$f(t) = \begin{cases} \frac{1}{\beta} e^{-t/\beta} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

where $\beta < \alpha$. What happens to the response as $\beta \rightarrow 0$? Comment briefly on your result. [15]